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NURTURING VALUES AND EXCELLENCE

PROJECT SYNOPSIS

PROJECT TITLE –

InternSync – Internship Management and Evaluation Platform

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1. INTRODUCTION

InternSync is a full-stack web-based Internship Management System developed to simplify communication, task tracking, submission validation, and performance evaluation between interns and mentors. In many organizations, managing interns through emails and messaging applications can lead to miscommunication, missed deadlines, and poor documentation.

InternSync solves this problem by providing a centralized platform with role- based access and secure authentication. The system supports structured task management, automated report validation (such as page-count verification), attendance tracking, leave management, and marks-based performance evaluation. It replaces informal communication methods with a professional and well-documented digital workflow.

With increasing digital transformation, organizations are adopting automated and cloud-based management systems that provide centralized access and real- time updates. InternSync follows this approach by integrating automation and structured workflow management to improve efficiency, transparency, and accountability throughout the internship process.

2. WHAT IS THE PROJECT ABOUT?

InternSync is a web-based Internship Workflow and Evaluation Management System developed to manage internship activities through a centralized platform. It improves communication, task management, report submission, attendance tracking, leave management, and performance evaluation between interns and mentors. The system replaces informal communication methods like emails or messaging apps with a secure and structured environment that ensures transparency and proper documentation throughout the internship process.

The system uses role-based access where users register and log in as either an Intern or a Mentor, and a unique Intern ID is generated during registration.

After login, users are directed to their respective dashboards. Interns can view assigned tasks, deadlines, and feedback, submit project reports with GitHub repository links, and apply for leave while tracking approval status. The system automatically validates report submissions by checking requirements such as page count and records submission time. Mentors can assign tasks, review submissions, evaluate intern performance, manage attendance, approve or reject leave requests, and publish announcements. All data is securely stored with authentication to ensure authorized access only.

The system is intended for interns, mentors, companies, educational institutions, and training organizations that need a structured way to manage internship programs.

InternSync follows a three-tier architecture. The frontend is developed using HTML, CSS (Tailwind), and JavaScript for the user interface. The backend is built with Node.js and Express.js, handling authentication, role-based access, and business logic. The system runs on Node.js and can be deployed on cloud platforms like Render for remote access and scalability.

3. PROBLEM STATEMENT

- Internship management often relies on manual methods (emails, chats, spreadsheets, paper records).
- Leads to miscommunication between interns and mentors.
- No centralized system → data scattered and hard to manage.
- Difficult to track multiple interns and maintain records efficiently.
- No automated validation → unverified submissions and inconsistent documentation.
- Manual attendance & leave tracking → prone to errors and data loss.
- Performance evaluation is unstructured and often subjective.
- Overall need for a centralized, secure, automated system for better communication, tracking, and accuracy.

4. WHAT PROBLEM DOES IT SOLVE?

InternSync addresses these challenges by providing a centralized, role- based web platform that digitizes and automates the entire internship workflow. The system eliminates dependency on informal communication channels by offering structured dashboards for interns and mentors.

It introduces automated report validation, including page-count verification, deadline tracking, and timestamp recording, ensuring submission accuracy and transparency. Attendance and leave management are digitized, reducing manual errors and improving record maintenance. The marks-based evaluation system standardizes performance assessment using defined scoring parameters such as punctuality, efficiency, and understanding.

Compared to traditional methods, InternSync improves security through authenticated access, enhances transparency through real-time tracking, reduces administrative workload through automation, and increases overall efficiency in managing multiple interns. The system ultimately benefits both mentors and interns by creating a professional, accountable, and organized internship work management environment.

5. OBJECTIVES

The primary objective is to design and develop an efficient, centralized, and automated internship management system to streamline processes and improve coordination.

- Develop a centralized internship management platform
- Automate task assignment, submission tracking, and report validation
- Implement automated document verification (e.g., 100-page report check)
- Provide a structured, marks-based performance evaluation system
- Digitize attendance and leave management processes
- Ensure secure, role-based authentication and data access
- Reduce manual errors and improve workflow efficiency
- Enhance transparency and accountability between interns and mentors

6. SCOPE

InternSync covers the complete internship lifecycle within an organization. The system includes:

- Role-based authentication for Interns and Mentors
- Task creation, assignment, and deadline tracking
- Word document submission with automated page-count validation
- GitHub repository link submission
- Attendance management system
- Leave request and approval workflow
- Marks-based evaluation and feedback system
- Private mentor-intern communication
- Announcement and notification system
- Secure MySQL-based data storage

Exclusions -

- AI-based plagiarism detection
- Integration with external HR or payroll systems
- Multi-organization (multi-company) deployment in a single system
- Native mobile application (web-based only)

Limitations -

- The system does not include real-time video conferencing or virtual meeting integration; external platforms must be used for live meetings.
- Page validation checks quantity (page count) but not content quality.
- Requires stable internet connectivity for real-time access.
- Designed primarily for small to medium-scale internship programs.

7. WHERE IT CAN BE USED?

- InternSync can be implemented in various real-world environments where structured internship or training management is required. It is suitable for:
- IT companies and software development firms managing technical interns
- Corporate organizations conducting training or management trainee programs
- Educational institutions coordinating industrial internships for students
- Startups handling project-based internship programs
- Training institutes and skill development centers
- Research organizations supervising academic interns
- The system is particularly useful in environments where multiple interns are handled simultaneously and structured monitoring, evaluation, and documentation are essential.

8. PROGRAMMING LANGUAGE, TOOLS, DATABASE & FRAMEWORK

Programming Languages:

- HTML
- CSS (Tailwind CSS)
- JavaScript
- JavaScript (Node.js)
- Python

Framework / Technologies:

- Express.js – Backend web framework for Node.js
- JWT (JSON Web Token) – Authentication & role-based access control
- Multer – File upload handling (report submissions)
- REST-based client-server architecture
- Tailwind CSS – Frontend styling framework

Database:

- MySQL – Relational Database Management System

Tools Used:

- Visual Studio Code – IDE
- Node.js Runtime Environment – Server-side execution

Hosting Platform:

- Render – Cloud-based deployment platform used to host the web application online, enabling remote access, scalability, and real-time availability

9. SYSTEM FUNCTIONALITY

InternSync consists of the following major modules and features:

- Authentication & User Management -

  Secure role-based registration and login (Intern/Mentor) with unique Intern ID generation, email-password authentication, session management, and profile management including editing details, photo upload, and account tracking.

- Intern Dashboard - View assigned tasks, deadlines, and reference reports; submit Word reports with GitHub links; track submission history, attendance summary, leave requests and approval status; receive notifications and communicate privately with mentors.
- Mentor Dashboard - Create and assign tasks with deadlines and requirements; review submissions with automatic validation results; evaluate interns using marks-based criteria; manage attendance, approve or reject leave requests, publish announcements, and control project access duration.
- Submission & Validation System - Upload Word documents with file-format restrictions, automatic page- count verification, and timestamp-based deadline validation.
- Notification & Communication System - Automated alerts for task assignments, km submissions, feedback, announcements, and leave approval or rejection updates.
- Search & Access Management - Mentors can quickly search and access intern profiles using unique Intern IDs.

10.HOW THE SYSTEM WORKS

- The user registers and logs into the system using secure credentials.
- The system verifies the user role (Intern or Mentor).
- Based on the role, the respective dashboard is loaded.

For Intern:

- The intern views assigned tasks, deadlines, and model report.
- The intern uploads the project report (Word document) and GitHub repository link.
- The system performs automatic page-count validation.
- Submission timestamp is recorded and stored in the MySQL database.

For Mentor:

- The mentor creates and assigns tasks to interns.
- The mentor reviews submitted projects and checks validation results.
- The mentor evaluates performance using marks-based scoring criteria.
- Attendance is marked and leave requests are approved or rejected.
- Feedback and notifications are sent to the intern.
- Provide Certificate for intern.
- All data transactions are processed through the backend server and securely stored in the MySQL database. The system ensures role-based access, structured workflow management, and real-time updates.

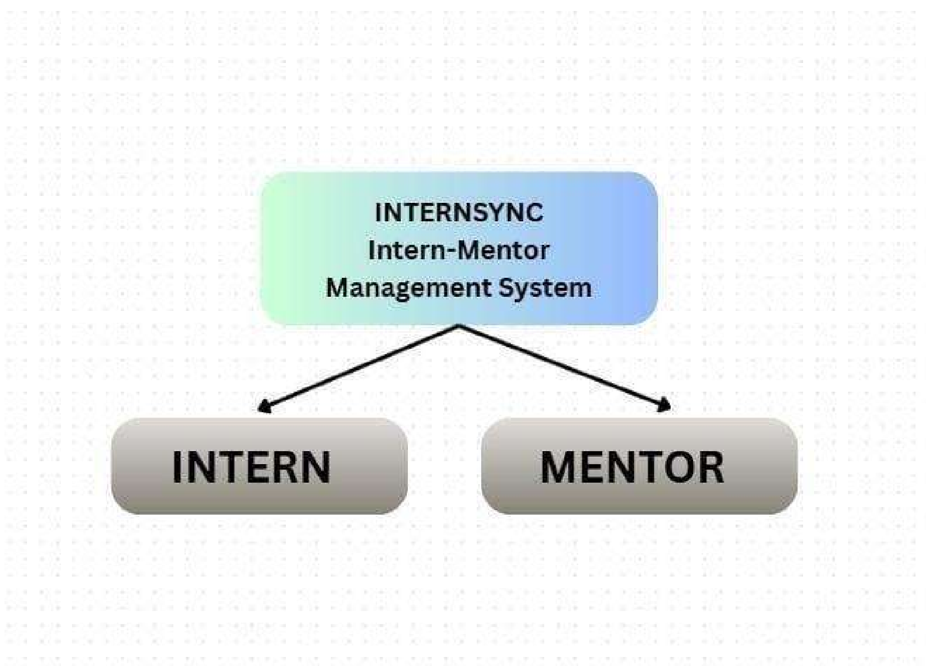

fig 10.1 - Login modes

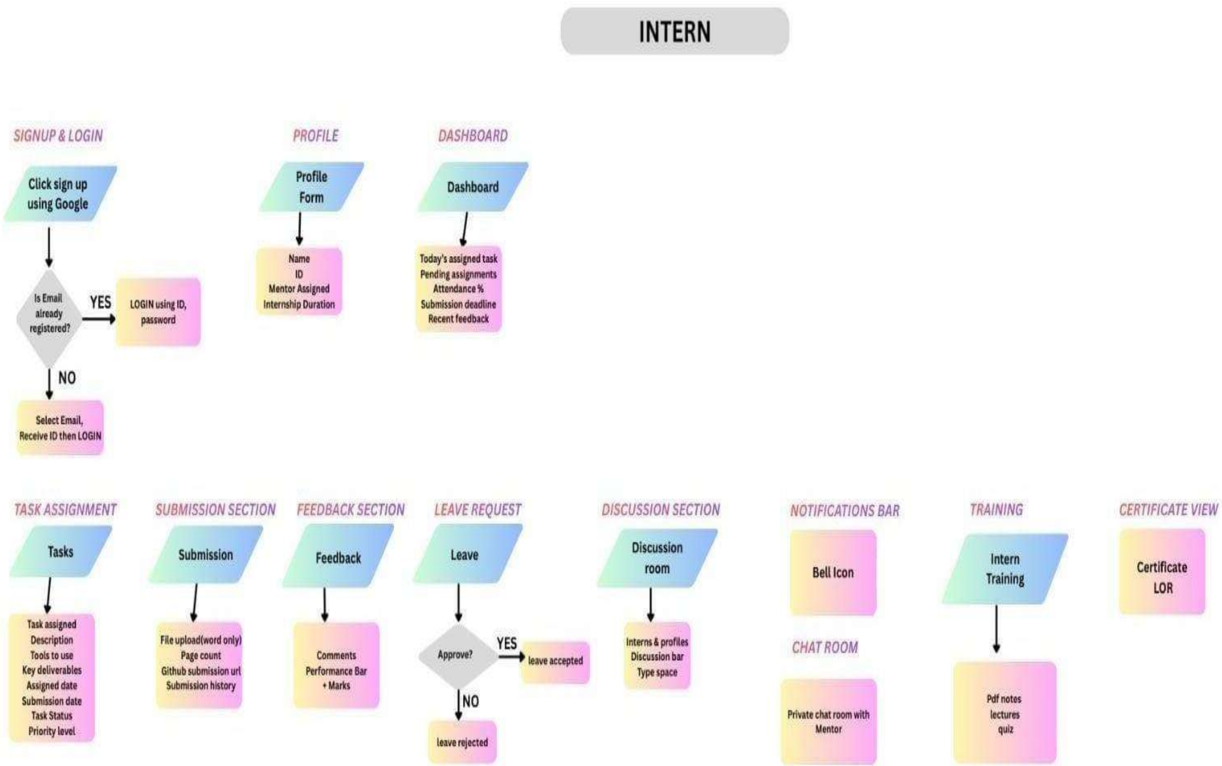

fig 10.2 - Intern mode

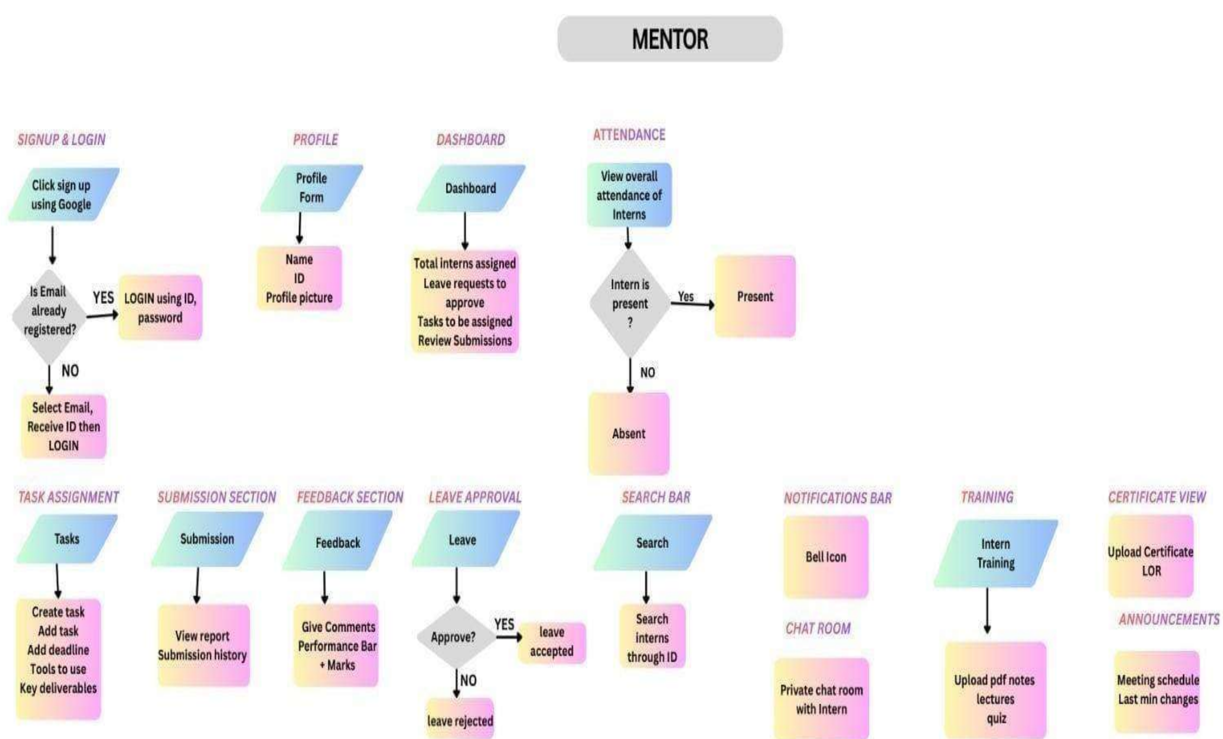

fig 10.3 - Mentor mode

11.HARDWARE & SOFTWARE REQUIREMENTS

- Hardware Requirements:
  Processor: Intel i3 or higher
  RAM: Minimum 4 GB (8 GB recommended) Storage:
  Minimum 20 GB free disk space

- Software Requirements:
  Operating System: Windows / Linux / macOS

  IDE: Visual Studio Code

  Browser: Google Chrome (for testing and debugging)

  Database Server: MySQL Server

  Backend Runtime: Node.js (for server-size logic)

  Hosting Platform: Render (Cloud-based deployment platform)

12.EXPECTED OUTCOME

The implementation of InternSync is expected to deliver a structured and centralized internship management system that enhances organizational efficiency and transparency.

The system will:

- Improve efficiency by automating task tracking, submission validation, and attendance management.
- Increase accuracy by reducing manual errors in record maintenance.
- Save time through automated page-count validation and deadline tracking
- Standardize performance evaluation using marks-based scoring criteria.
- Enhance data security through role-based authentication and controlled access.
- Improve user satisfaction by providing a clear, organized, and professional workflow.

Overall, the system ensures better coordination between interns and mentors while maintaining proper documentation and accountability.

13.CONCLUSION

InternSync is a comprehensive web-based Internship Management System designed to digitize and automate the internship lifecycle within organizations. The system integrates role-based authentication, task management, automated report validation, attendance tracking, leave management, performance evaluation, and communication features into a single centralized platform.

By replacing manual and informal processes with a structured digital solution, InternSync enhances efficiency, transparency, and reliability in internship programs. The project demonstrates the practical application of full-stack web development concepts and modern deployment practices using cloud hosting.

In conclusion, InternSync provides a scalable, secure, and professional solution that improves internship workflow management and contributes to organizational productivity and structured evaluation systems.